

Smart Traffic Lights Using Local Artificial Intelligence

Basheer

Department of Computer Science and Engineering

Ballari institute of technology and management

Ballari, India

mubasheer@gmail.com

Abstract—Smart traffic-light systems are increasingly being investigated to improve traffic management by responding to changing road conditions. Conventional traffic signals generally operate according to predefined timing plans and may not respond efficiently to sudden variations in traffic demand. Artificial intelligence (AI), computer vision, and edge computing provide alternative approaches in which traffic conditions can be detected and processed closer to the intersection. This paper presents a conceptual smart traffic-light architecture based on local AI processing. Street cameras capture traffic information, a local computing device analyzes the information, and the traffic-signal controller uses the resulting traffic information for adaptive operation. A cloud server may optionally be used for data storage and large-scale analysis. The proposed architecture also considers emergency-vehicle signal preemption. Unlike the unsupported assumption that emergency-vehicle path clearing is generally unavailable, emergency vehicle preemption is already an established traffic-management technology. Therefore, the research opportunity is to investigate the integration of reliable local computer vision with existing traffic signal control and emergency-vehicle preemption mechanisms.

Index Terms—Smart traffic lights, artificial intelligence, edge computing, computer vision, adaptive traffic signal control, emergency vehicle preemption, intelligent transportation systems.

I. INTRODUCTION

Traffic congestion is an important challenge in modern urban transportation systems. Traffic signals are used to regulate vehicle movements at intersections and to coordinate traffic flows. Traditional traffic signals can operate using fixed timing plans or coordinated timing schedules. These approaches are relatively simple and predictable, but their performance may decrease when traffic demand differs significantly from the conditions for which the timing plans were designed [13], [14].

Adaptive traffic signal control provides an alternative approach by using traffic measurements to modify signal timing according to current traffic conditions. Research has investigated several artificial intelligence and machine-learning methods for traffic signal control, including deep reinforcement learning [1], [6].

Computer vision is another important technology for intelligent transportation systems. Camera-based systems can estimate traffic information such as vehicle presence, vehicle count, and traffic flow. Machine-vision-based approaches have been investigated for adaptive traffic-light control [2], [9].

Edge computing provides an additional architectural approach. Instead of sending all camera data to a remote cloud

server, some processing can be performed near the intersection. This can reduce the amount of raw video transmitted over a network and can allow the system to make use of locally available traffic information [3].

This paper proposes a conceptual local-AI architecture for smart traffic lights. The architecture combines street cameras, local computing, traffic-signal control, and optional cloud connectivity. Emergency vehicle preemption is also considered as an additional component of the system.

II. BACKGROUND AND RELATED WORK

A. Traditional Traffic Signal Control

Traditional traffic signals commonly use predetermined timing plans. The duration of green, yellow, and red phases can be configured according to expected traffic conditions. Such systems can be effective when traffic patterns are relatively stable. However, unexpected changes in traffic demand may result in inefficient use of green time.

Adaptive signal control attempts to overcome this limitation by using traffic information to modify signal timing. The Federal Highway Administration describes adaptive signal control as a technology that uses current traffic information to adjust signal timing [13].

B. Artificial Intelligence for Traffic Signals

Artificial intelligence has been investigated for adaptive traffic signal control. Deep reinforcement learning is particularly relevant because an agent can learn signal-control policies through interaction with a traffic environment. Rasheed *et al.* reviewed deep reinforcement learning approaches for traffic signal control and discussed several research challenges [1].

Other studies have investigated reinforcement-learning approaches for cooperative and adaptive traffic signal control [4], [5]. These approaches demonstrate the potential of learning-based methods, although practical deployment requires consideration of safety, computational requirements, training data, and communication infrastructure.

C. Computer Vision

Computer vision can provide traffic information using road-side cameras. Vehicle detection and tracking can be used to estimate traffic characteristics. Gorodokin *et al.* investigated machine vision for adaptive traffic-light control [2].

YOLO-based object detection has also been investigated for traffic monitoring and traffic signal applications [9]. The accuracy of a camera-based system, however, depends on factors such as camera position, illumination, weather conditions, image quality, and the selected detection model.

D. Edge Computing

Cloud computing can provide substantial computational resources for AI-based traffic analysis. However, transmitting continuous camera streams to a remote server requires network connectivity and can increase communication requirements.

Edge computing provides an alternative in which processing is performed closer to the data source. A local computer installed near a traffic intersection can process camera information and produce traffic measurements locally. Research has investigated smart traffic monitoring using computer vision and edge computing [3].

Local processing does not automatically guarantee a specific response time or cost. Actual performance depends on the hardware, AI model, software implementation, camera characteristics, and communication architecture.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed architecture consists of four major components:

- 1) **Street Cameras:** Cameras capture images or video of traffic approaching an intersection.
- 2) **Smart Pole Computer:** A local computing device processes camera data using a computer-vision model. It can estimate vehicle count, traffic density, queue length, or vehicle presence.
- 3) **Traffic Signal Controller:** The processed traffic information can be provided to the signal controller. The controller remains responsible for implementing signal timing and safety logic.
- 4) **Cloud Server:** A cloud server can optionally store traffic information and support long-term analysis, monitoring, or model development.

The basic architecture can be represented as

Street Cameras → Smart Pole Computer → Traffic Signal Controller. (1)

An optional cloud connection can be represented as

Smart Pole Computer → Cloud Server. (2)

The local computer performs the initial processing, while the cloud server is used for optional storage and large-scale analysis.

IV. WORKING PRINCIPLE

The proposed system operates in several stages.

First, street cameras continuously capture traffic conditions near an intersection. The captured images are provided to a local computer.

Second, the local computer applies a computer-vision model to identify vehicles and estimate traffic conditions. Possible outputs include vehicle count, queue length, traffic density, and vehicle type.

Third, the traffic information is provided to the traffic-signal controller. Based on the selected control strategy, the signal timing can be adjusted according to the detected traffic conditions.

Fourth, selected traffic information can be transmitted to a cloud server. The cloud server can be used for historical analysis, system monitoring, model training, and network-level traffic analysis.

This architecture avoids the requirement that every camera frame must be transmitted to a remote server for processing.

V. EMERGENCY VEHICLE SIGNAL PREEMPTION

Emergency-vehicle signal preemption is an established traffic signal control technology. It allows normal signal operation to be temporarily modified to provide priority to an approaching emergency vehicle [15], [16].

Therefore, the research gap should not be described as the absence of automatic emergency-vehicle path clearing. Instead, an important research problem is the integration of local computer vision with existing emergency-vehicle preemption systems.

A local AI system could potentially detect an emergency vehicle using camera information and generate an appropriate request to the traffic signal controller. For a route containing multiple intersections, the system could investigate how preemption requests should be coordinated between intersections.

Emergency vehicle signal control must also consider false detections, detection latency, conflicting requests, pedestrian safety, traffic queues, and the requirements of the traffic-signal controller [11], [12].

VI. COMPARISON OF TRAFFIC SIGNAL APPROACHES

The three approaches considered in this paper are conventional traffic-signal control, cloud-based AI processing, and local AI processing.

TABLE I

CONCEPTUAL COMPARISON OF TRAFFIC SIGNAL APPROACHES

Characteristic	Conventional	Local AI
Traffic input	Timing plans	Camera/sensor data
Adaptation	Limited	Possible
Local processing	Low	Required
Network dependence	Application dependent	Can be reduced
AI capability	Not required	Possible
Emergency preemption	Possible	Can be integrated
Cloud processing	Not required	Optional

Specific values for latency, installation cost, processing capacity, and network consumption should be determined

through experimental measurements rather than presented as universal values.

VII. RESEARCH GAP

Existing research demonstrates the use of adaptive traffic signal control, machine vision, reinforcement learning, and edge computing for intelligent transportation applications [1]–[3], [6].

Emergency vehicle signal preemption is also an established technology and has been investigated in the context of intelligent transportation systems [11], [12].

Based on these developments, a suitable research direction is the development and evaluation of a local-AI traffic signal system that combines traffic monitoring with emergency-vehicle detection and preemption.

The following areas can be investigated:

- Accurate emergency-vehicle detection using roadside cameras.
- Low-latency AI inference using local computing hardware.
- Reliable communication between the AI system and traffic signal controller.
- Coordination of emergency-vehicle priority across multiple intersections.
- Reduction of false-positive and false-negative detections.
- Comparison between local and cloud-based processing.
- Evaluation of traffic performance under different traffic conditions.

VIII. CHALLENGES

Although local AI provides several potential advantages, practical implementation presents several challenges.

A. Detection Accuracy

Camera-based detection may be affected by poor lighting, rain, fog, occlusion, and camera placement. These conditions can reduce the accuracy of vehicle and emergency-vehicle detection.

B. Computational Requirements

Local AI requires sufficient processing capability to run computer vision models. The hardware must provide an appropriate balance between computational performance, energy consumption, and cost.

C. Safety

Traffic signals are safety-critical systems. An AI system should not directly override safety mechanisms without appropriate validation and control logic.

D. Communication

Although local processing can reduce dependence on cloud communication, communication with the traffic-signal controller may still be necessary. Reliable communication is therefore important.

E. False Detection

A false emergency-vehicle detection could unnecessarily interrupt normal traffic. Conversely, failure to detect an emergency vehicle could prevent the intended priority operation. These cases should be considered during system evaluation.

IX. FUTURE WORK

Future work can involve implementing the proposed architecture using a roadside camera, an edge-computing device, and a computer-vision model. A suitable object-detection model can be trained or configured to identify vehicles and emergency vehicles.

The system can initially be evaluated in a simulation environment. Traffic simulators can be used to compare conventional signal timing with adaptive control and emergency-vehicle priority strategies.

After simulation, controlled real-world testing can be considered. Important performance measures include vehicle delay, queue length, travel time, emergency-vehicle travel time, detection accuracy, false positive rate, false negative rate, and processing latency.

X. CONCLUSION

This paper presented a conceptual architecture for smart traffic lights using local artificial intelligence. The architecture uses roadside cameras, local AI processing, a traffic-signal controller, and optional cloud connectivity.

Existing research demonstrates that machine vision, reinforcement learning, adaptive signal control, and edge computing can contribute to intelligent transportation systems. However, performance claims such as fixed processing times, installation costs, or computational capacity should be supported by experimental measurements.

Emergency-vehicle signal preemption should also be considered an existing traffic-management capability rather than an entirely new function. The research opportunity is to investigate how local computer vision can be integrated with traffic-signal control and existing emergency-vehicle preemption mechanisms in a reliable and safe manner.

REFERENCES

- [1] F. Rasheed, K. L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, “Deep reinforcement learning for traffic signal control: A review,” *IEEE Access*, vol. 8, pp. 208016–208044, 2020, doi: 10.1109/ACCESS.2020.3034141.
- [2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, “Optimization of adaptive traffic light control modes based on machine vision,” *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021, doi: 10.1016/j.trpro.2021.09.047.
- [3] G. Liu, H. Shi, A. Kiani, A. Khreishah, J. Lee, N. Ansari, C. Liu, and M. M. Yousef, “Smart traffic monitoring system using computer vision and edge computing,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 12027–12038, 2022, doi: 10.1109/TITS.2021.3109481.
- [4] M. Wang, L. Wu, J. Li, and L. He, “Traffic signal control with reinforcement learning based on region-aware cooperative strategy,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 6774–6785, 2022, doi: 10.1109/TITS.2021.3062072.
- [5] D. Ma, B. Zhou, X. Song, and H. Dai, “A deep reinforcement learning approach to traffic signal control with temporal traffic pattern mining,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 11789–11800, 2022, doi: 10.1109/TITS.2021.3107258.

- [6] M. Miletić, E. Ivanjko, M. Gregurić, and K. Kušić, "A review of reinforcement learning applications in adaptive traffic signal control," *IET Intelligent Transport Systems*, vol. 16, no. 10, pp. 1269–1285, 2022, doi: 10.1049/itr2.12208.
- [7] P. Jing, H. Huang, and L. Chen, "An adaptive traffic signal control in a connected vehicle environment: A systematic review," *Information*, vol. 8, no. 3, Art. no. 101, 2017, doi: 10.3390/info8030101.
- [8] Y. Wang and X. Yang, "A review of the self-adaptive traffic signal control system based on future traffic environment," *Journal of Advanced Transportation*, vol. 2018, Art. no. 1096123, 2018, doi: 10.1155/2018/1096123.
- [9] K. Zaatouri and T. Ezzedine, "A self-adaptive traffic light control system based on YOLO," in *Proc. 2018 Int. Conf. Internet of Things, Embedded Systems and Communications (IINTEC)*, 2018, pp. 16–19, doi: 10.1109/IINTEC.2018.8695293.
- [10] S. Liu, G. Wu, and M. J. Barth, "A complete state transition-based traffic signal control using deep reinforcement learning," in *Proc. 2022 IEEE Conf. Technologies for Sustainability (SusTech)*, 2022, pp. 100–107, doi: 10.1109/SusTech53338.2022.9794168.
- [11] M. Cao, Q. Shuai, and V. O. K. Li, "Emergency vehicle-centered traffic signal control in intelligent transportation systems," in *Proc. 2019 IEEE Intelligent Transportation Systems Conf. (ITSC)*, 2019, pp. 4525–4531, doi: 10.1109/ITSC.2019.8916879.
- [12] Y.-Y. Chen, J.-Y. Wang, S.-C. Lo, and W.-T. Sung, "An emergency vehicle traffic signal preemption system considering queue spillbacks along routes and negative impacts on non-priority traffic," *IET Intelligent Transport Systems*, vol. 18, no. 8, pp. 1385–1395, 2024, doi: 10.1049/itr2.12518.
- [13] Federal Highway Administration, "Adaptive signal control technology," U.S. Department of Transportation. [Online]. Available: <https://www.fhwa.dot.gov/innovation/everydaycounts/edc-1/asct.cfm>
- [14] Federal Highway Administration, "Model systems engineering documents for adaptive signal control technology (ASCT) systems," U.S. Department of Transportation. [Online]. Available: https://ops.fhwa.dot.gov/publications/fhwahop11027/sec_d.htm
- [15] Federal Highway Administration, "Traffic signal timing manual: Chapter 9 – Advanced signal timing topics," U.S. Department of Transportation. [Online]. Available: <https://ops.fhwa.dot.gov/publications/fhwahop08024/chapter9.htm>
- [16] Federal Highway Administration, "Signalized intersections: Informational guide – Signal preemption and/or priority," U.S. Department of Transportation. [Online]. Available: <https://www.fhwa.dot.gov/publications/research/safety/04091/08.cfm>